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1. $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, $\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$

By Newton's Law,

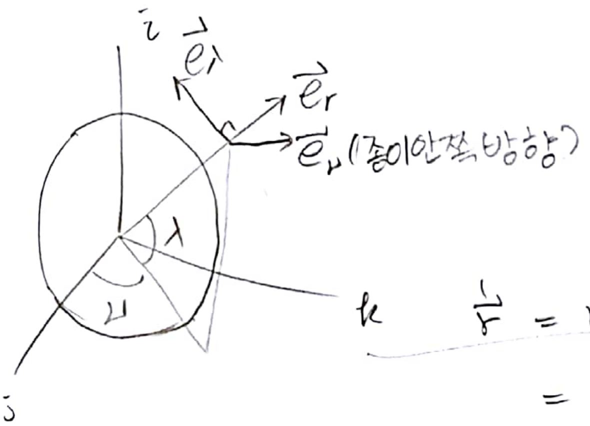
$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{F} \Rightarrow \begin{cases} \ddot{x} = -\frac{\mu}{r^3}x + F_x \\ \ddot{y} = -\frac{\mu}{r^3}y + F_y \\ \ddot{z} = -\frac{\mu}{r^3}z + F_z \end{cases}$$

where $r = \sqrt{x^2+y^2+z^2}$

m? −2.

$$\Rightarrow \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \\ \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\mu}{r^3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{\mu}{r^3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\frac{\mu}{r^3} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}.$$

3. Let $\vec{r} = r\vec{e}_r$, $\vec{F} = F_r\vec{e}_r + F_\nu\vec{e}_\nu + F_\lambda\vec{e}_\lambda$



$$\Rightarrow \vec{\omega} = \dot{\nu}\cos\lambda\,\vec{e}_\lambda + \dot{\nu}\sin\lambda\,\vec{e}_r - \dot{\lambda}\vec{e}_\nu$$

$$\begin{cases} \vec{e}_r = \vec{e}_\nu \times \vec{e}_\lambda \\ \vec{e}_\nu = \vec{e}_\lambda \times \vec{e}_r \\ \vec{e}_\lambda = \vec{e}_r \times \vec{e}_\nu \end{cases}$$

$$\dot{\vec{r}} = \dot{r}\vec{e}_r + r\dot{\vec{e}}_r = \dot{r}\vec{e}_r + r\vec{\omega}\times\vec{e}_r \text{ (by Coriolis thm)}$$
$$= \dot{r}\vec{e}_r + r\dot{\nu}\cos\lambda\,\vec{e}_\lambda\times\vec{e}_r - r\dot{\lambda}\vec{e}_\nu\times\vec{e}_r$$
$$= \dot{r}\vec{e}_r + r\dot{\nu}\cos\lambda\,\vec{e}_\nu + r\dot{\lambda}\vec{e}_\lambda$$

$$\ddot{\vec{r}} = \ddot{r}\vec{e}_r + (\dot{r}\dot{\nu}\cos\lambda + r\ddot{\nu}\cos\lambda - r\dot{\nu}\dot{\lambda}\sin\lambda)\vec{e}_\nu + (\dot{r}\dot{\lambda} + r\ddot{\lambda})\vec{e}_\lambda$$
$$+ \dot{r}(\vec{\omega}\times\vec{e}_r) + r\dot{\nu}\cos\lambda(\vec{\omega}\times\vec{e}_\nu) + r\dot{\lambda}(\vec{\omega}\times\vec{e}_\lambda)$$
$$= \ddot{r}\vec{e}_r + (\dot{r}\dot{\nu}\cos\lambda + r\ddot{\nu}\cos\lambda - r\dot{\nu}\dot{\lambda}\sin\lambda)\vec{e}_\nu + (\dot{r}\dot{\lambda} + r\ddot{\lambda})\vec{e}_\lambda$$
$$+ (\dot{r}\dot{\nu}\cos\lambda\,\vec{e}_\lambda - \dot{r}\dot{\lambda}\vec{e}_\nu)\times\vec{e}_r$$
$$+ (r\dot{\nu}^2\cos^2\lambda\,\vec{e}_\lambda + r\dot{\nu}^2\cos\lambda\sin\lambda\,\vec{e}_r)\times\vec{e}_\nu$$
$$+ (r\dot{\lambda}\dot{\nu}\sin\lambda\,\vec{e}_r - r\dot{\lambda}^2\vec{e}_\nu)\times\vec{e}_\lambda$$

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$= \vec{e}_r(\ddot{r} - r\dot{\nu}^2\cos^2\lambda - r\dot{\lambda}^2) + \vec{e}_\lambda(\dot{r}\dot{\lambda} + r\ddot{\lambda} + \dot{r}\dot{\lambda} + r\dot{\nu}^2\cos\lambda\sin\lambda)$

$+(\dot{r}\dot{\nu}\cos\lambda + r\ddot{\nu}\cos\lambda - r\dot{\nu}\dot{\lambda}\sin\lambda + \dot{r}\dot{\nu}\cos\lambda - r\dot{\lambda}\dot{\nu}\sin\lambda)\vec{e}_\nu$

$= (\ddot{r} - r\dot{\nu}^2\cos^2\lambda - r\dot{\lambda}^2)\vec{e}_r + (r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2\cos\lambda\sin\lambda)\vec{e}_\lambda$

$+(r\ddot{\nu}\cos\lambda + 2\dot{r}\dot{\nu}\cos\lambda - 2r\dot{\nu}\dot{\lambda}\sin\lambda)\vec{e}_\nu$

Since $\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{F} = -\frac{\mu}{r^2}\vec{e}_r + \vec{F}$, then

−2

$$\begin{cases} \ddot{r} - r\dot{\nu}^2\cos^2\lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + F_r \checkmark \text{ m?} \\ r\ddot{\nu}\cos\lambda + 2\dot{r}\dot{\nu}\cos\lambda - 2r\dot{\nu}\dot{\lambda}\sin\lambda = F_\nu \\ r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2\cos\lambda\sin\lambda = F_\lambda \end{cases}$$

C. Let $\bar{r} = R$, $\bar{\nu} = \sqrt{\frac{\mu}{R^3}}\,t = \omega t$, $\bar{\lambda} = 0$.

$\begin{cases} \sin\Delta\theta \approx \Delta\theta \\ \cos\Delta\theta \approx 1 \end{cases}$

Define $r = R + \Delta r$, $\nu = \omega t + \Delta\nu$, $\lambda = \Delta\lambda$.

From $\ddot{r} - r\dot{\nu}^2\cos^2\lambda - r\dot{\lambda}^2 + \frac{\mu}{r^2} = F_r$,

$\Rightarrow (\cancel{\ddot{R}}^0 + \Delta\ddot{r}) - (R+\Delta r)(\omega + \Delta\dot{\nu})^2\underbrace{\cos^2\Delta\lambda}_{\approx 1} - (R+\Delta r)\cancel{\Delta\dot{\lambda}^2}^0 + \frac{\mu}{(R+\Delta r)^2}$

$= \Delta\ddot{r} - (R+\Delta r)(\omega^2 + 2\omega\Delta\dot{\nu} + \cancel{\Delta\dot{\nu}^2}^0) + \frac{\mu}{R^2}\left(1 + \frac{\Delta r}{R}\right)^{-2}$

Taylor expansion OR Binomial expansion

$(\because)\left|\frac{\Delta r}{R}\right| \ll 1$

$(1+\alpha x)^{-a} \approx 1 - a\alpha x$

$= \Delta\ddot{r} - R\omega^2 - 2\omega R\Delta\dot{\nu} - \omega^2\Delta r - \cancel{2\omega\Delta\dot{\nu}\Delta r}^0 + \frac{\mu}{R^2}\left(1 - 2\frac{\Delta r}{R}\right)$ $\quad \frac{\mu}{R^2} = \omega^2 R$

$= \Delta\ddot{r} - \cancel{\omega^2 R} - 2R\omega\Delta\dot{\nu} - \omega^2\Delta r + \cancel{\omega^2 R} - 2\omega^2\Delta r$

$= \Delta\ddot{r} - 2R\omega\Delta\dot{\nu} - 3\omega^2\Delta r.$ V m? −2.

$\therefore \Delta\ddot{r} - 2R\omega\Delta\dot{\nu} - 3\omega^2\Delta r = F_r.$

From $r\ddot{\nu}\cos\lambda + 2\dot{r}\dot{\nu}\cos\lambda - 2r\dot{\nu}\dot{\lambda}\sin\lambda = F_\nu$,

$\Rightarrow (R+\Delta r)\Delta\ddot{\nu}\underbrace{\cos\Delta\lambda}_{\approx 1} + 2(\cancel{\dot{R}}^0 + \Delta\dot{r})(\omega + \Delta\dot{\nu})\underbrace{\cos\Delta\lambda}_{\approx 1} - 2(R+\Delta r)(\omega + \Delta\dot{\nu})\Delta\dot{\lambda}\underbrace{\sin\Delta\lambda}_{\approx \Delta\lambda}$

$= (R+\Delta r)\Delta\ddot{\nu} + 2\Delta\dot{r}(\omega + \Delta\dot{\nu}) - 2(R+\Delta r)(\omega + \Delta\dot{\nu})\cancel{\Delta\lambda^2}^0$

$= R\Delta\ddot{\nu} + \cancel{\Delta r\Delta\ddot{\nu}}^0 + 2\omega\Delta\dot{r} + \cancel{2\Delta\dot{r}\Delta\dot{\nu}}^0$

$= R\Delta\ddot{\nu} + 2\omega\Delta\dot{r}$

$R\Delta\ddot{\nu} + 2\omega\Delta\dot{r} = F_\nu.$

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③

From $r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2 \cos\lambda \sin\lambda = F_\lambda$,

$\Rightarrow (R+\Delta r)\Delta\ddot{\lambda} + 2(\cancel{\dot{R}}^{0} + \Delta\dot{r})\Delta\dot{\lambda} + (R+\Delta r)(\omega + \Delta\dot{\nu})^2 \underbrace{\cos\Delta\lambda}_{\approx 1} \underbrace{\sin\Delta\lambda}_{\approx \Delta\lambda}$

$= R\Delta\ddot{\lambda} + \Delta r \cancel{\Delta\ddot{\lambda}}^{0} + 2\Delta\dot{r}\cancel{\Delta\dot{\lambda}}^{0} + (R+\Delta r)(\omega^2 + 2\Delta\dot{\nu}\omega + \cancel{\Delta\dot{\nu}^2}^{0})\Delta\lambda$

$= R\Delta\ddot{\lambda} + (R+\Delta r)(\omega^2 + 2\Delta\dot{\nu}\omega)\Delta\lambda$

$= R\Delta\ddot{\lambda} + R\omega^2\Delta\lambda + \cancel{\Delta r\omega^2\Delta\lambda}^{0} + \cancel{2R\omega\Delta\dot{\nu}\Delta\lambda}^{0} + \cancel{\Delta r 2\Delta\dot{\nu}\omega\Delta\lambda}^{0}$

$= R\Delta\ddot{\lambda} + \omega^2 R\Delta\lambda$

$\therefore \underline{R\Delta\ddot{\lambda} + \omega^2 R\Delta\lambda = F_\lambda}$

Let $x = \Delta r$, $y = R\Delta\nu$, $z = R\Delta\lambda$, then

$$\begin{cases} \ddot{x} - 2\omega\dot{y} - 3\omega^2 x = F_r \\ \ddot{y} + 2\omega\dot{x} = F_\nu \\ \ddot{z} + \omega^2 z = F_\lambda \end{cases}$$

which are CW equations.

#2.

A. Let $\omega_x = 0 = \omega_y$ & $\omega_z = \overline{\omega_z} = \text{constant}$, $M_x = 0 = M_y = M_z$.

$I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x$

$= (I_y - I_z)\, 0 \cdot \overline{\omega_z} + 0 = 0$

$I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y$

$= (I_z - I_x)\, 0 \cdot \overline{\omega_z} + 0 = 0$.

$I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z$

$= (I_x - I_y)\, 0 \cdot 0 + 0 = 0$.

$$\Rightarrow \begin{cases} I_x\dot{\omega}_x = 0 \\ I_y\dot{\omega}_y = 0 \\ I_z\dot{\omega}_z = 0 \end{cases}$$

I_x, I_y, I_z가 0인 경우는 물리적으로 부피가 있는 물체에서는 나올 수 없으므로 고려하지 않음

Since $\boxed{I_x > 0, I_y > 0, I_z > 0}$ by definition, then

$\dot{\omega}_x = 0 = \dot{\omega}_y = \dot{\omega}_z$

Thus it is an equilibrium solution.

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3. Let $\omega_x = \Delta\omega_x$, $\omega_y = \Delta\omega_y$, $\omega_z = \bar{\omega}_z + \Delta\omega_z$.

From $I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + M_x$,

$$\Rightarrow M_x = I_x\dot{\omega}_x - (I_y - I_z)\omega_y\omega_z$$

$$= I_x\Delta\dot{\omega}_x - (I_y - I_z)\Delta\omega_y(\bar{\omega}_z + \Delta\omega_z)$$

$$= I_x\Delta\dot{\omega}_x - (I_y - I_z)\Delta\omega_y\bar{\omega}_z - (I_y - I_z)\cancel{\Delta\omega_z\Delta\omega_z}^{0}$$

$$= I_x\Delta\dot{\omega}_x - (I_y - I_z)\bar{\omega}_z\Delta\omega_y.$$

From $I_y\dot{\omega}_y = (I_z - I_x)\omega_x\omega_z + M_y$,

$$\Rightarrow M_y = I_y\dot{\omega}_y - (I_z - I_x)\omega_x\omega_z.$$

$$= I_y\Delta\dot{\omega}_y - (I_z - I_x)\Delta\omega_x(\bar{\omega}_z + \Delta\omega_z)$$

$$= I_y\Delta\dot{\omega}_y - (I_z - I_x)\Delta\omega_x\bar{\omega}_z - (I_z - I_x)\cancel{\Delta\omega_x\Delta\omega_z}^{0}$$

$$= I_y\Delta\dot{\omega}_y - (I_z - I_x)\bar{\omega}_z\Delta\omega_x.$$

From $I_z\dot{\omega}_z = (I_x - I_y)\omega_x\omega_y + M_z$,

$$\Rightarrow M_z = I_z\dot{\omega}_z - (I_x - I_y)\omega_x\omega_y$$

$$= I_z(\cancel{\dot{\bar{\omega}}_z}^{0} + \Delta\dot{\omega}_z) - (I_x - I_y)\cancel{\Delta\omega_x\Delta\omega_y}^{0}.$$

$$= I_z\Delta\dot{\omega}_z.$$

$$\therefore \begin{cases} I_x\Delta\dot{\omega}_x - (I_y - I_z)\bar{\omega}_z\Delta\omega_y = M_x. \\ I_y\Delta\dot{\omega}_y - (I_z - I_x)\bar{\omega}_z\Delta\omega_x = M_y. \\ I_z\Delta\dot{\omega}_z = M_z. \end{cases}$$

$\to$ Coupled (first two equations)

$\leftarrow$ Decoupled (third equation).

c.

$$\begin{cases} \Delta\dot{\omega}_x = \dfrac{I_y - I_z}{I_x}\bar{\omega}_z\Delta\omega_y + \dfrac{M_x}{I_x}. \\ \Delta\dot{\omega}_y = \dfrac{I_z - I_x}{I_y}\bar{\omega}_z\Delta\omega_x + \dfrac{M_y}{I_y} \\ \Delta\dot{\omega}_z = \dfrac{M_z}{I_z} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \Delta\dot{\omega}_x \\ \Delta\dot{\omega}_y \\ \Delta\dot{\omega}_z \end{bmatrix} = \begin{bmatrix} 0 & \dfrac{I_y - I_z}{I_x}\bar{\omega}_z & 0 \\ \dfrac{I_z - I_x}{I_y}\bar{\omega}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\omega_x \\ \Delta\omega_y \\ \Delta\omega_z \end{bmatrix} + \begin{bmatrix} M_x/I_x \\ M_y/I_y \\ M_z/I_z \end{bmatrix}$$

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$$\begin{vmatrix} \lambda & -\frac{I_y - I_z}{I_x}\bar{\omega}_z & 0 \\ -\frac{I_z - I_x}{I_y}\bar{\omega}_z & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} = (-1)^{3+3}\lambda \begin{vmatrix} \lambda & -\frac{I_y - I_z}{I_x}\bar{\omega}_z \\ -\frac{I_z - I_x}{I_y}\bar{\omega}_z & \lambda \end{vmatrix}$$

By Clamer's rule.

$$= \underbrace{\lambda\left(\lambda^2 + \frac{(I_y - I_z)(I_x - I_z)}{I_x I_y}\bar{\omega}_z^2\right) = 0.}_{\text{Characteristic equation.}}$$

① If I_z is major ($I_z > I_y$ & $I_z > I_x$) OR minor ($I_z < I_y$ & $I_z < I_x$), then eigenvalues are

$$\lambda = 0, \ \pm j\bar{\omega}_z\sqrt{\frac{(I_y - I_z)(I_x - I_z)}{I_x I_y}}.$$

Hence the system is marginary stable. because there exist one zero and two pure imaginary.

② If I_z is Intermediate, ($I_x < I_z < I_y$ OR $I_y < I_z < I_x$) then eigenvalues are

$$\lambda = 0, \ \pm \bar{\omega}_z\sqrt{\frac{(I_y - I_z)(I_z - I_x)}{I_x I_y}}$$

Thus the ~~system~~ is unstable because ~~there exists~~ a positive real eigenvalue.

③ If $I_z = I_x$ OR $I_z = I_y$, then all eigenvalues are zero.

$$\lambda = 0.\ 0.\ 0.$$

Then the system is unstable because there are many zero eigenvalues.

($I_x = I_y \neq I_z$인 경우는 ①에, $I_x = I_y = I_z$인 경우는 ②에 포함)

이지아

20210032 (partially visible)

$I = \mathrm{diag}(200, 200, 200)\ kg\,m^2$. $\vec{a} = \begin{bmatrix} 0.424 \\ -0.566 \\ 0.707 \end{bmatrix}$

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rest to rest. $\therefore \dot{\phi}_0 = 0 = \dot{\phi}_f$.

$R_0 = \exp[S(a)] \Rightarrow \phi_0 = 1$. $\because |\vec{a}| = \sqrt{0.424^2 + (-0.566)^2 + (0.707)^2} = 1$.

$R_f = I = \exp[0] \Rightarrow \phi_f = 0$.

Assume $\ddot{\phi}(t) = \begin{cases} -0.01\ \text{rad/sec} & \text{for } 0 \le t \le t_s \\ 0.01\ \text{rad/sec} & \text{for } t_s < t \le t_f \end{cases}$

From $R = \exp(S(\vec{a})\phi)$, $\dot{R} = S(\vec{a})\dot{\phi}(t)\exp(S(\vec{a})\phi) = S(\vec{a})\dot{\phi}(t)R$.

Since $\dot{R} = S(\vec{\omega})R$ where $S(\cdot): \mathbb{R}^3 \to \mathbb{R}^{3\times3}$ is skew-symmetric mapping, then $S(\vec{\omega})R = \dot{R} = S(\vec{a})\dot{\phi}(t)R = S(\vec{a}\dot{\phi})R$.

$$\therefore \vec{\omega} = \vec{a}\dot{\phi}$$

If $0 \le t \le t_s$,

$$\dot{\phi}(t) = \int_0^t \ddot{\phi}(\tau)\,d\tau + \dot{\phi}(0)$$

$$= \int_0^t -0.01\,d\tau + 0$$

$$= -0.01t.$$

Thus $\dot{\phi}(t_s) = -0.01t_s$.

If $t_s < t \le t_f$,

$$\dot{\phi}(t) = \int_0^t \ddot{\phi}(\tau)d\tau + \dot{\phi}(0)$$

$$= \int_0^{t_s} \ddot{\phi}(\tau)\,d\tau + \int_{t_s}^{t} \ddot{\phi}(\tau)d\tau + \dot{\phi}(0)$$

$$= -0.01t_s + \int_{t_s}^{t} 0.01\,d\tau + 0$$

$$= -0.01t_s + 0.01(t - t_s) = 0.01(t - 2t_s)$$

$$\therefore \vec{\omega} = \vec{a}\dot{\phi}(t) = \begin{cases} -0.01t\,\vec{a} & \text{for } 0 \le t \le t_s \\ 0.01(t-2t_s)\,\vec{a} & \text{for } t_s < t \le t_f. \end{cases}$$

$\vec{\omega}(0) = 0 = \vec{\omega}(t_f)$. ($\because$ rest to rest maneuver)

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From Euler's rotational equations,

$$\vec{M} = I\dot{\vec{\omega}} + \vec{\omega} \times I\vec{\omega}$$

$$= I\vec{a}\ddot{\phi}(t) + \vec{a}\dot{\phi}(t) \times I\vec{a}\dot{\phi}(t).$$

$$= 200\ddot{\phi}(t)\vec{a} + 200\dot{\phi}^2(t)\,\vec{a}\times\vec{a} = 200\ddot{\phi}(t)\vec{a}$$

Hence control moment vector is,

$$\vec{M} = \begin{cases} -2\vec{a} & \text{for } 0 \le t \le t_s \\ 2\vec{a} & \text{for } t_s < t \le t_f. \end{cases}$$

By the boundary condition, $\vec{\omega}(0) = 0 = \vec{\omega}(t_f)$.

$\vec{\omega}(0) = -0.01 \times 0 \times \vec{a} = 0.$

$\vec{\omega}(t_f) = 0.01\,(t_f - 2t_s)\,\vec{a} = 0.$

Since $\vec{a} \neq 0$, $t_f - 2t_s = 0.$ $\therefore t_s = \frac{t_f}{2}$

If $0 \le t \le t_s = t_f/2$,

$$\phi(t) = \int_0^t \dot{\phi}(\tau)\,d\tau + \phi(0)$$

$$= \int_0^t (-0.01\tau)\,d\tau + 1.$$

$$= -0.005t^2 + 1.$$

Then $\phi(t_s) = \phi(t_f/2) = -0.005\left(\frac{t_f}{2}\right)^2 + 1 = -\frac{1}{800}t_f^2 + 1.$

If $t_s = t_f/2 < t \le t_f$,

$$\phi(t) = \int_0^t \dot{\phi}(\tau)\,d\tau + \phi(0)$$

$$= \int_0^{t_f/2} (-0.01\tau)\,d\tau + \int_{t_f/2}^{t} 0.01(\tau - t_f)\,d\tau + 1.$$

$$= -\frac{1}{800}t_f^2 + \int_{t_f/2}^{t} 0.01\tau\,d\tau - \int_{t_f/2}^{t} 0.01\,t_f\,d\tau + 1.$$

$$= -\frac{1}{800}t_f^2 + \frac{1}{200}\left(t^2 - \frac{t_f^2}{4}\right) - \frac{1}{100}t_f\left(t - \frac{t_f}{2}\right) + 1$$

Then $\phi(t_f) = -\frac{1}{800}t_f^2 + \frac{1}{200}\left(t_f^2 - \frac{t_f^2}{4}\right) - \frac{1}{100}t_f\left(t_f - \frac{t_f}{2}\right) + 1$

$$= -\frac{1}{800}t_f^2 + \frac{t_f^2}{200}\cdot\frac{3}{4} - \frac{t_f^2}{100}\cdot\frac{1}{2} + 1.$$

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$$= -\frac{1}{800}t_f^2 + \frac{3}{800}t_f^2 - \frac{1}{200}t_f^2 + 1$$

$$= \frac{-1+3-4}{800}t_f^2 + 1$$

$$= -\frac{1}{400}t_f^2 + 1.$$

Since $\phi(t_f) = 0$, $-\frac{1}{400}t_f^2 + 1 = 0$. $\Rightarrow t_f^2 - 400.$

Since $t_f > 0$, $t_f = 20$ sec.

4. $I_y \Delta\ddot{\theta} = 3\omega_0^2(-I_x + I_z)\Delta\theta + M_y$.

$$\Rightarrow 200\,\Delta\ddot{\theta} = 3\omega_0^2(-150 + 100)\Delta\theta + M_y$$

$$\Rightarrow \Delta\ddot{\theta} = -\frac{3}{4}\omega_0^2\Delta\theta + \frac{M_y}{200}.$$

$$\Rightarrow \begin{bmatrix}\Delta\dot{\theta}\\ \Delta\ddot{\theta}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{3}{4}\omega_0^2 & 0\end{bmatrix}\begin{bmatrix}\Delta\theta\\ \Delta\dot{\theta}\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}\frac{M_y}{200}.$$

$\Delta\theta_0 = 45° = \pi/4$ rad. $\Delta\theta_f = 0° = 0$ rad.

rest to rest $\Rightarrow \Delta\dot{\theta}_0 = 0 = \Delta\dot{\theta}_f$

Let assume that t_s is the time that the single impulse is applied.

Then if $0 \le t \le t_s$, the spacecraft move without external moment.

$$\Rightarrow \begin{bmatrix}\Delta\dot{\theta}(t)\\ \Delta\ddot{\theta}(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ \frac{4}{3}\omega_0^2 & 0\end{bmatrix}\begin{bmatrix}\Delta\theta(t)\\ \Delta\dot{\theta}(t)\end{bmatrix}.$$

Let $a^2 = \frac{4}{3}\omega_0^2$; $a > 0$. ($a = \frac{2\sqrt{3}}{3}\omega_0 \approx 1.3856\text{e-}03$ rad/sec).

$$\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix} = \begin{bmatrix}-a^2 & 0\\ 0 & -a^2\end{bmatrix} = -a^2 I.$$

$$\Rightarrow \begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}^3 = -a^2\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}$$

$$\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}^4 = -a^2\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}^2 = (-a^2)^2 I.$$

$$\Rightarrow \exp\left(\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}t\right) = I + \begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}t + \frac{1}{2!}\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}^2 t^2 + \cdots$$

$$= I\left(1 + \frac{1}{2!}(-a^2)t^2 + \frac{1}{4!}(-a^2)^2t^4 + \cdots\right)$$

$$+ \begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}\left(t + \frac{1}{3!}(-a^2)t^3 + \frac{1}{5!}(-a^2)^2t^5 + \cdots\right)$$

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$$= I\left(\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!}(at)^{2k}\right) + \frac{1}{a}\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}\left(\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{(2k-1)!}(at)^{2k-1}\right).$$

$$= I\cos(at) + \begin{bmatrix}0 & a^{-1}\\ -a & 0\end{bmatrix}\sin(at)$$

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$$= \begin{bmatrix}\cos at & \sin at/a\\ -a\sin at & \cos at\end{bmatrix}.$$

$$\Rightarrow \begin{bmatrix}\Delta\theta(t)\\ \Delta\dot\theta(t)\end{bmatrix} = \exp\left(\begin{bmatrix}0 & 1\\ -a^2 & 0\end{bmatrix}t\right)\begin{bmatrix}\Delta\theta(0)\\ \Delta\dot\theta(0)\end{bmatrix}$$

$$= \begin{bmatrix}\cos at & \sin at/a\\ -a\sin at & \cos at\end{bmatrix}\begin{bmatrix}\pi/4\\ 0\end{bmatrix}.$$

$$\Rightarrow \begin{cases}\Delta\theta(t_s) = \frac{\pi}{4}\cos at_s.\\ \Delta\dot\theta(t_s) = -\frac{\pi}{4}a\sin at_s.\end{cases}$$

Change in the pitch rate due to impulse

추력을 준 직후, when $t = t_s^+$

$$\Delta\theta(t_s^+) = \frac{\pi}{4}\cos at_s \quad \& \quad \Delta\dot\theta(t_s^+) = -\frac{\pi}{4}a\sin at_s + \Delta\dot\theta_s$$

최종 시간 t_f 에서의 상태는,

$$\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}\Delta\theta(t_f)\\ \Delta\dot\theta(t_f)\end{bmatrix} = \underbrace{\begin{bmatrix}\cos a(t_f-t_s) & \sin a(t_f-t_s)/a\\ -a\sin a(t_f-t_s) & \cos a(t_f-t_s)\end{bmatrix}}_{STM}\begin{bmatrix}\frac{\pi}{4}\cos at_s\\ -\frac{\pi}{4}a\sin at_s + \Delta\dot\theta_s\end{bmatrix}$$

STM의 determinant를 구해보면 $\cos^2 a(t_f-t_s) + \sin^2 a(t_f-t_s) = 1$

이므로 $\ker(STM) = \{0\}$ 이다.

따라서 최종 상태 ($\Delta\theta_f = 0$ & $\Delta\dot\theta_f = 0$)를 만족하는 $\Delta\theta(t_s^+)$, $\Delta\dot\theta(t_s^+)$가 유일하게 결정된다

따라서 추력은 마지막 순간에 줘야 한다. ($t_s = t_f$). (처이 마지막)

$\Delta\theta_f = 0$ & $\Delta\dot\theta_f = 0$ 이므로 $\Delta\theta(t_s^+) = 0$ & $\Delta\dot\theta(t_s^+) = 0$ 이어야 한다?

$$\Rightarrow \frac{\pi}{4}\cos at_s = 0. \quad \& \quad -\frac{\pi}{4}a\sin at_s + \Delta\dot\theta_s = 0$$

$$\Rightarrow at_s = \left(\frac{2n-1}{2}\right)\pi \quad \text{for } n\in\mathbb{Z}^+\cup\{0\}$$

$$\Rightarrow t_s = 1133.6246 + 2267.2492n \text{ sec for } n\in\mathbb{Z}^+\cup\{0\}$$

If n is even or 0, then $\Delta\dot\theta_s = \frac{\pi}{4}a = 1.0883e-03$ rad/sec

If n is odd, then $\Delta\dot\theta_s = -\frac{\pi}{4}a = -1.0883e-03$ rad/sec.

Change in the pitch rate due to impulse

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Thus rest to rest maneuver can be performed using a single impulsive if given transfer time is greater than 1133.6246 sec. open 5/5

The impulse should be performed at t_s, and it is same as transfer time.

ideal 천이시간을 설정하는 경우 특정 시간 ($1133.6246(2n-1)$ $n \in \mathbb{Z}^+$) 동안만 위성이 이동한다.

⇒ Change in the pitch rate due to impulse 10/10.

$$\Delta \dot{\theta}_s = \begin{cases} 1.0883e{-03} \text{ rad/sec} & \text{if } n \equiv 0 \pmod 2 \\ -1.0883e{-03} \text{ rad/sec} & \text{if } n \equiv 1 \pmod 2 \end{cases}$$

where $n \in \mathbb{Z}^+ \cup \{0\}$.

~~Transfer time — impulse 사용 시간.~~

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